Autonomous Smart Caddy Cart (ASCC)

ELEC 490/498 Engineering Blueprint

ASCC project logo
Replace with team-designed logo

Group 07
Vince Song (21vs51)
Connor Gorman (21crg6)
JP Masotti (21jpm28)

Supervisor: Prof. Karen Rudie
Instructor: Prof. Muhammad Alam
Industry collaborator: None specified

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Executive Summary

Walking golfers currently use separate equipment to transport clubs, track club inventory, and obtain distance information. The ASCC prototype combines these tasks in one motorized push cart. It will follow one golfer at a controlled distance, stop in response to obstacles or a wireless emergency-stop command, detect removal and return of tagged clubs, and show GPS-derived distance to a green on an onboard display.

Development proceeds through early drive and tag-sensing proofs of concept, independent following and safety prototypes, a display/GPS prototype, integrated cart operation, and outdoor verification and validation. The intended operating scenario is a single golfer on open, dry grass, with an operator supervising moving tests. Shot tracking, automated club recommendations, wet-weather operation, and multi-person following are outside the committed scope. The estimated project cost is CAD 595, including contingency, against a CAD 600 budget. The design will be judged using the traceable requirements and acceptance tests defined in this blueprint.

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Acronyms

ASCC: Autonomous Smart Caddy Cart
GPS: Global Positioning System
RFID: Radio Frequency Identification
V&V: Verification and Validation
MVP: Minimum Viable Product
UWB: Ultra-Wideband
GPIO: General-Purpose Input/Output

1. Introduction

1.1 Problem and motivation

Commercial following carts reduce the need to push golf equipment but do not inherently provide bag inventory or green-distance information. Separate golf-tracking systems provide some of these functions, often through additional wearable or mobile devices **stewart**, **arccos**, **arccoswatch**. Prior work surveys person-following technologies and their tradeoffs [1]; field studies highlight measurement limitations in golf GPS tracking [2]. ASCC combines transport and on-cart information rather than attempting automatic shot logging.

1.2 Objective, scope and operational scenario

The stakeholder is a walking golfer who wants a phone-free round. The system boundary includes the motorized chassis, control electronics, sensors, tagged-club monitoring, GPS and display. External interactions include a golfer carrying a tracking device, terrain and obstacles, satellite signals, green-coordinate data, and the operator's wireless emergency stop. The planned demonstration uses one golfer, one bag, open dry grass and supervised operation. The cart reports distance from its own position, not the ball or golfer. Wet surfaces, steep slopes, crowded areas and autonomous course navigation are excluded.

1.3 System context

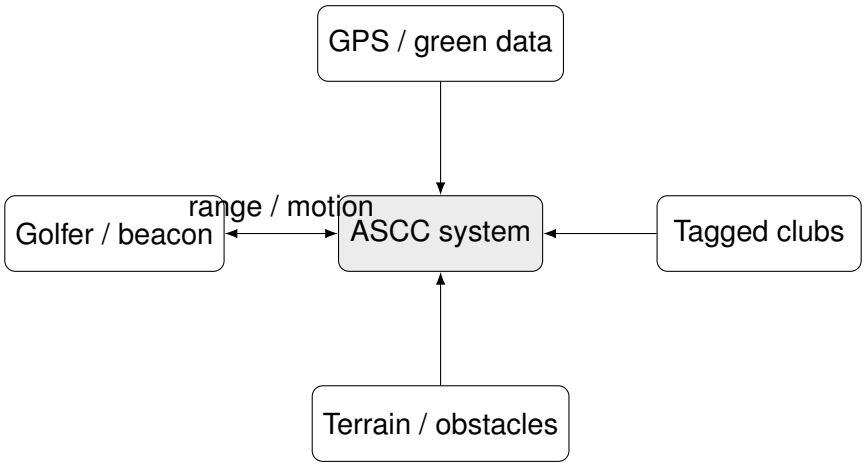


Figure 1: ASCC context and external information flows (original schematic).

1.4 Scope exclusions

Shot classification, club recommendations, mobile-app integration and autonomous navigation to waypoints are stretch possibilities only; none are required for MVP acceptance.

2. Functional Description and System Design

2.1 System decomposition and interfaces

The system is divided into five logical subsystems: (1) golfer ranging and following control, (2) obstacle and emergency-stop safety, (3) drive and power, (4) RFID inventory, and (5) GPS, green-data and display. The Raspberry Pi coordinates high-level logic and display; a dedicated motor-control/safety path should ensure that loss of high-level commands does not cause continued motion. Exact processor, ranging radio, motor driver and RFID-reader interfaces remain to be confirmed by component selection.

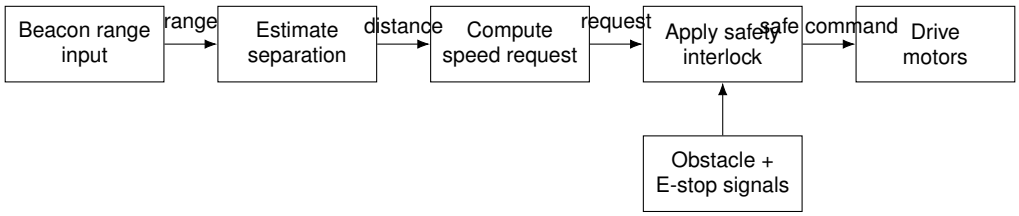


Figure 2: Fully decomposed example of the following-control function, including safety interlock.

2.2 Functional-to-logical allocation

Table 1: Function allocation and interface labels.

Function	Logical element	Interface / requirement
Estimate target range	Ranging receiver	Beacon measurements; R-IF01
Compute motion request	Following controller	Distance error; R-F01
Inhibit unsafe motion	Safety controller	Obstacle/E-stop; R-S01–R-S03
Audit club slots	RFID reader / inventory service	Tag events; R-C01
Calculate green range	GPS/display service	Position + map data; R-G01

2.3 Design choices and alternatives

Sensor-based tracking avoids camera lighting and training dependencies, but reliable target identification must be demonstrated experimentally. RFID tag placement and reader geometry will be tested against NFC or per-slot detection if bulk reads fail. GPS green-distance display is simpler than shot tracking but depends on valid green coordinates. A separate safety inhibit is preferred over relying only on the display computer; wiring and fail-safe behavior must be checked during prototype development.

2.4 Identified prototypes

P01 drive/power bench rig; P02 RFID bag-slot rig; P03 following/ranging rig; P04 obstacle/E-stop rig; P05 GPS/display rig; P06 integrated stationary cart; P07 moving outdoor MVP. Each has a test and milestone in Sections 5–6.

3. Requirements Specifications

3.1 Stakeholder needs and hierarchy

N01: transport a bag while following one walking golfer. N02: prevent unintended cart motion. N03: identify missing clubs. N04: display distance to green without a phone. N05: remain feasible under budget and schedule. These needs flow into system-level functions (R-F, R-S, R-C, R-G), and then measurable interface/performance requirements (R-IF, R-P).

Table 2: Critical system-level requirements and traceability.

ID / priority	Requirement statement	Need	Test
R-F01 / Critical	The cart shall follow one designated golfer at a target separation of 1.5 m with tolerance ± 0.3 m in the defined test scenario.	N01	T03
R-S01 / Critical	The cart shall inhibit drive commands within 300 ms of detecting a test obstacle; 200 ms is the design target.	N02	T04
R-S02 / Critical	The cart shall stop or disable motor drive within 100 ms after receipt of a valid emergency-stop command; 50 ms is the design target.	N02	T05
R-S03 / Critical	The cart shall not continue commanded autonomous motion after target-tracking signal loss beyond a validated timeout.	N02	T06
R-C01 / Major	The inventory system shall report present/absent status for 14 designated tagged club slots.	N03	T07
R-G01 / Major	The display shall show the distance from cart GPS position to the selected green with error no greater than 5 m under the defined reference test.	N04	T08
R-P01 / Major	The cart shall limit maximum autonomous cruise speed to 5.0 km/h ± 0.5 km/h during acceptance testing.	N01,N02	T03

3.2 Interface, performance and fault requirements

Table 3: Additional verifiable requirements.

ID	Requirement statement	Test
R-IF01	The tracking link shall support a nominal 15 m range (± 2 m) in an unobstructed test area.	T02
R-IF02	The wireless E-stop link shall operate at 20 m minimum range (30 m design target).	T05
R-IF03	The onboard display shall have at least 5 in diagonal size (7 in design target).	T08
R-P02	The cart shall cover 8.5 km or more on one charge under the defined loaded test (10 km design target).	T09
R-S04	The cart shall not energize the drive when E-stop is latched or when controller power is reset.	T05
R-C02	The inventory system shall issue a missing-club warning when a previously detected club is absent during an audit.	T07
R-M01	The prototype shall be constructed within a CAD 600 component budget, counting the planned procurement items.	T10
R-X01 / Stretch	The system may log individual shots for future club recommendations.	Not committed

3.3 Critical requirement rationale

R-S01 and R-S02 are safety-critical because a moving loaded cart could contact a golfer or obstacle. Their acceptance measurements refer to detection/command-to-drive-inhibit latency, not stopping distance; braking distance must be recorded separately during T04/T05. R-S03 requires the team to select and document a safe timeout before the test; this parameter is deliberately unresolved. R-G01 requires known reference coordinates and a documented GPS test environment. The 14-slot assumption and range/speed targets originate in the proposal and remain subject to feasibility validation.

4. Risks and Safety Hazards

Risks are assessed provisionally on likelihood (L) and severity (S) scales from 1 (low) to 5 (high); score = L$\times$S. These are planning estimates, not measured probabilities. A relevant laboratory electrical safety procedure and component datasheets must be cited and reviewed before energizing hardware. For moving tests, the safety operator retains a

Table 4: Initial project risk register (team to validate).

Hazard / failure	L	S	Score	Mitigation and affected work
Unintended cart movement	3	5	15	Independent inhibit, supervised tests; P01/P04, M04–M08
RFID misses club tags	4	3	12	Reader-placement trials and per-slot fallback; P02, M02
Loss of golfer tracking	3	5	15	Signal-loss stop and range checks; P03/P04
Battery short or overheating	2	5	10	Fuse, enclosure, cable protection; P01
GPS error or unavailable green data	4	2	8	Reference survey, fallback warning; P05
Cart sourcing delay	3	4	12	Alternative chassis and early procurement; M01

physical means of stopping the cart, and tests are suspended for unsafe slopes, pedestrians, poor visibility or moisture. The team must document battery chemistry-specific handling and the university's applicable SOP before hardware trials.

5. Development Approach and Milestones

An incremental, risk-first approach progresses from bench prototypes to integrated outdoor trials. Subsystem communication formats, power requirements and fault states are checked before integration.

Table 5: Blueprint milestone plan (proposed; confirm with team).

ID	Target date	Deliverable / evidence	Lead	Prototype/test
M01	Oct 30, 2026	Chassis and drive control demonstrator	Connor	P01/T01
M02	Nov 13, 2026	RFID tag/slot feasibility	JP	P02/T07
M03	Nov 20, 2026	Pi sensor and interface bring-up	Vince	P03/T02
M04	Dec 04, 2026	Following control on bench / low-speed rig	Vince	P03/T03
M05	Dec 18, 2026	Club inventory integrated with bag	JP	P02/T07
M06	Jan 22, 2027	Obstacle and emergency-stop inhibit	Vince	P04/T04–T06
M07	Feb 05, 2027	GPS and status display	JP	P05/T08
M08	Feb 19, 2027	Stationary system integration and interlocks	All	P06/T05–T08
M09	Mar 01, 2027	Controlled moving system integration	All	P07/T03–T06
M10	Mar 15, 2027	Outdoor validation and range test	All	P07/T09–T10

5.1 Prototype purpose and exit gates

P01 validates safe wheel actuation without autonomous control; P02 establishes reliable tagged-club detection; P03 establishes target ranging and a bounded speed request; P04 verifies independent safety overrides; P05 checks GPS-to-green calculations and display; P06 validates interface compatibility without moving; P07 verifies the complete operating scenario. A prototype advances only after its linked acceptance test passes or a documented corrective action is accepted.

5.2 Contingencies and dependencies

P01 is prerequisite for moving P03/P04 trials. P04 safety acceptance precedes P07. If RFID reliability fails, reposition antennas or use slot-level sensing; this affects M05 but not safe mobility. If target ranging fails, restrict motion to controlled manual/remote operation while redesigning P03; do not claim autonomous-following completion. If GPS accuracy fails, report unavailable/unreliable distances rather than false precision. Preserve M09–M10 integration buffer by deferring stretch features.

6. Verification and Validation Testing

Verification compares measurements against requirements. Validation evaluates whether a golfer can use the cart for the intended phone-free scenario. A safety operator is present for all powered-motion trials. Measurement instruments, number of repeats, weather,

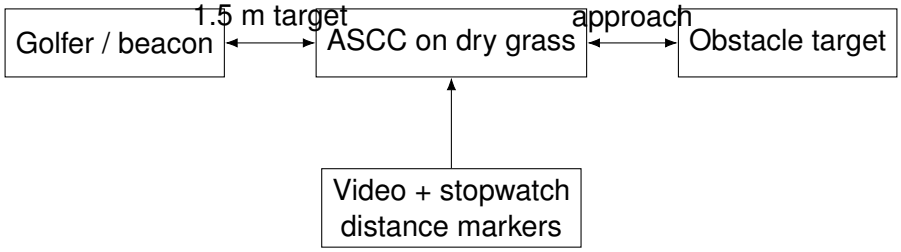


Figure 3: Outdoor following and obstacle-response experimental arrangement (not to scale).

Table 6: Acceptance tests and linked prototype/requirement.

Test	Prototype	Procedure / evidence	Pass condition
T01	P01	Lift wheels, issue stop/start, log direction and motor state	Correct commands, safe reset
T02	P03	Measure beacon reception at 13, 15, 17 m	Document achieved range vs R-IF01
T03	P03/P07	Mark following separation and speed over repeated grass runs	1.5 ± 0.3 m; 5 ± 0.5 km/h
T04	P04/P07	Trigger obstacle input, record sensor-to-inhibit latency on video/log	$\leq$300 ms
T05	P04/P06	Trigger E-stop at 20 m; time drive disable and power-reset behavior	$\leq$100 ms; no restart
T06	P03/P04	Deliberately disconnect target signal during low-speed trial	Safe stop within documented time-out
T07	P02	Remove/replace each of 14 tagged clubs; compare inventory	All slots report correct binary state
T08	P05	Compare displayed green distance against reference at surveyed points	$\leq$5 m error; display $\geq$5 in
T09	P07	Log loaded travel distance on one charge	$\geq$8.5 km
T10	P07	Reconcile procurement receipts and validate phone-free use on one hole	$\leq$CAD 600; core functions usable

terrain and test logs will be recorded. The T02 acceptance protocol must be refined because the original 15 ± 2 m proposal value is a nominal target rather than an explicit minimum. No test has been executed as part of this draft.

7. Resource Requirements

Procurement evidence required: Attach dated supplier URLs/quotes for every purchased item before submission; the original proposal lists estimates but does not establish

Table 7: Proposal-derived preliminary budget (CAD; supplier quotes pending).

Item	Cost	Relevant milestones
Used motorized cart	250	M01, M08–M10
Motor controller	50	M01, M06
Battery / power	75	M01–M10
Pi 5, ultrasonic sensors (owned)	0	M03–M10
LiDAR	50	M03, M06
RFID readers and tags	20	M02, M05
Mounting hardware	30	M08
Wiring, switches, connectors	30	M01–M08
Tax and shipping	75	All procurement
Contingency	15	All
Total	**595**	

current prices or vendor availability. The used cart listing may change, so record the selected listing and a fallback. Lab access, battery charging arrangements, test equipment and outdoor test permissions must be confirmed with the supervisor.

8. Conclusion

ASCC addresses the gap between following carts and separate golf-information devices by integrating transport, club inventory and green-distance display. The design is feasible only if the team demonstrates safe tracking and stopping before unsupervised-looking operation is attempted. The requirements-to-prototypes-to-tests chain establishes specific exit gates, while the CAD 595 estimate leaves only CAD 5 under the budget ceiling. The main limitations are outdoor tracking reliability, uncertain tag-read geometry, GPS error, and weather/terrain restrictions. Early prototypes should quantify these issues, and the team should narrow the operating envelope or redesign a subsystem if acceptance thresholds are not achieved. Current verified progress is limited to the proposal and the team's reported existing hardware; next steps are to confirm the chassis, finalize safety architecture, source components and run P01/P02 feasibility trials.

References

[1] T. E. D. Hall, A. G. B. Willmott, N. J. Hynes, J. E. T. Wells, and P. M. Allen, "The validity and reliability of two golf performance-tracking Global Positioning System (GPS) devices," *International Journal of Golf Science*, vol. 13, no. 1, pp. 103–121, 2025. [Online]. Available: https://www.golfsciencejournal.org/article/144587

[2] Stewart Golf. "Q Follow electric golf trolley," Accessed: Jul. 15, 2025. [Online]. Available: https://www.stewartgolf.co.uk/products/stewart-golf-q-follow

A. Detailed Traceability and Submission Checklist

A.1 Requirements hierarchy and critical relationship

N02 (avoid unintended motion) → safety inhibition → R-S01, R-S02, R-S03, R-S04 → P04 → T04, T05, T06 → M06 and M08. N03 (track clubs) → inventory audit → R-C01, R-C02 → P02 → T07 → M02/M05. This chain should be drawn as a dedicated vector hierarchy figure in the final submission.

A.2 Open decisions before submission

1. Confirm ranging technology, beacon interface, hardware fail-safe and motor controller ratings.
2. Confirm whether GPS coordinates of selected greens are available and licensed for demonstration.
3. Select RFID reader topology and document the 14-slot audit method.
4. Obtain component datasheets, relevant safety SOPs, and dated vendor quotations.
5. Replace cover placeholder with original ASCC logo; validate schedule with all team members.
6. Check body page count (Introduction through Conclusion must not exceed ten pages); shorten tables or move technical detail to this appendix as needed.
7. Review requirement acceptance definitions and sign off test repetitions, logging, and safe test conditions.